

Seunghwan Lee

Building human-centered interactive systems with AI, data, sensing, and prototyping

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Education

Mar. 2020 – Present **Chung-Ang University**
B.S. in Systems Biotechnology
Double Major in Photography
Minor in Nano-Convergence Materials Engineering

GPA: 4.24 / 4.5
Expected Graduation: Feb. 2027

Research Interests

- Human-Centered AI / AI × HCI
- Accessibility & Assistive Technology
- Digital Wellbeing & Behavior Change
- Multimodal & Physical Interaction
- Data-driven Interactive Systems

Selected Projects

I Can See 2 Jun. 2025 – Present

AI-Assisted Image-to-Braille & Audio Accessibility System

- Developed an AI-assisted accessibility system that transforms visual images into complementary Braille and audio descriptions for blind and low-vision users.
- Built an end-to-end pipeline integrating image capture, AI interpretation, Korean Braille conversion, tactile printing, and TTS, and refined the prototype through preliminary feedback from staff at the Siloam Center for the Blind.

Why do you read me? Sep. – Dec. 2025

LLM-Integrated Tangible Human-AI Interaction System

- Built a tangible AI system in which readers answer reflection questions through physical page navigation, with an LLM synthesizing their responses into personalized reflective feedback.
- Explored familiar physical interaction, such as page turning, as an alternative input modality for human-AI interaction beyond conventional screen- or chat-based interfaces.

I Need Rest

Mar. – Jun. 2025

Gamified Phone-Down System for Digital Wellbeing

- Designed and deployed a digital-wellbeing system that turns phone-down behavior into a measurable interaction using smartphone sensing, Firebase logging, and social feedback.
- Collected 87 rest records during a 10-day installation to explore voluntary phone-down behavior through lightweight sensing and gamified feedback.

City of Organism

Oct. – Dec. 2024

Public Data Processing & Data-Driven Visualization System

- Processed and mapped a month-long Seoul traffic dataset using road-link IDs, coordinates, lane counts, and hourly speed values.
- Joined multiple public datasets and reorganized them into coordinate-based time-series structures for dynamic 3D visualization in TouchDesigner.

Patents / Utility Models

Utility Model: Wearable Device Track Pad Using Infrared Light

Registration No. 20-0485654

- Registered an infrared-based on-body input concept that uses the thumbnail as an external trackpad for small-screen wearable interaction.

Utility Model: Stylus Pen

Registration No. 20-0485653

- Registered a battery-less stylus mechanism that uses piezoelectric elements to convert pen-tip pressure into magnetic signals for device-side pressure and tilt estimation.

Patent: Apparatus and Method for Unlocking Screen of Terminal

Registration No. 10-1825829

- Registered a mobile unlocking method that recognizes direction-vector sequences instead of fixed screen-point patterns.

Community Engagement

Portrait Photography for Elderly North Korean Defectors

Mar. – May 2026

Co-organizer & Volunteer Photographer, Gyeonggi Northwest Hana Center

8-hour event on May 8

- Co-organized and photographed formal portraits for elderly North Korean defectors living alone.
- Media coverage: Goyang Newspaper, “꽃그네 위에서 한 생애가 환하게 피어났다.”

Hana Family Kimchi-Making Event with North Korean Defectors

8 hours / Nov. 15, 2025

Event Support Volunteer, Gyeonggi Northwest Hana Center

- Supported on-site operations for an annual kimchi-making event for North Korean defectors, including participant guidance, ingredient and equipment preparation, distribution, and cleanup.

Color Purple

Sep. – Dec. 2024

Documentary Project on North Korean Defectors

- Documented the lived experiences, memories, and personal narratives of North Korean defectors in collaboration with Sae-teo Church in Yangcheon-gu.

Awards & Honors

AWARDS

Dec. 2024	Best Presentation, Group Category Open Studio, Department of Photography Project: City of Organism	Chung-Ang Univ.
Dec. 2024	Best Print, Individual Category Open Studio, Department of Photography Project: Color Purple	Chung-Ang Univ.

HONORS

Aug. 2025	Academic Excellence Scholarship	Chung-Ang Univ.
Aug. 2025	Microdegree Completion Merit Scholarship	Chung-Ang Univ.
Jul. 2025	Jeju Talent Development Scholarship	Jeju Inst. for Lifelong Education and Scholarship
Apr. 2025	CAU Career Development Scholarship	Chung-Ang Univ.
Mar. 2025	Extracurricular Achievement Scholarship	Chung-Ang Univ.
Feb. 2025	Academic Excellence Scholarship	Chung-Ang Univ.

Technical Training

AI Computer Vision Core Education Program

35 hours / Jan. 2026

Engineering Education Innovation Center, Chung-Ang University

- Studied computer vision fundamentals through lectures and Dofbot robotic arm practice; contributed to a team demo using YOLO-based object detection and camera-based robotic arm tracking.

Fusion 360 for 3D Design Education Program

48 hours / May – Sep. 2025

Engineering Education Innovation Center, Chung-Ang University

- Completed hands-on training in Fusion 360-based 3D modeling for digital fabrication and prototyping.

Leadership & Learning Activities

Generative AI Coaching Study

4 weeks / Oct. – Nov. 2024

Participant, NAVER Connect Foundation × MODULABS

- Participated in a team-based generative AI study using ChatGPT and Suno, producing music, images, videos, and documents through weekly missions.

AI Basics Coaching Study

5 weeks / Jan. – Feb. 2023

Team Leader, NAVER Connect Foundation × MODULABS

- Led weekly discussions on data analysis, Pandas, Matplotlib, PyTorch, and MNIST assignments; the team was selected as Outstanding Mission Team in Week 5.

Head First Rust Flipped Learning School

11 weeks / Nov. 2022 – Jan. 2023

Participant, MODULABS

- Participated in a peer-led Rust study and presented assigned chapters on collections, error handling, and concurrent programming.

PY4E: Python for Everybody Coaching Study

6 weeks / Jul. – Aug. 2022

Team Leader, NAVER Connect Foundation × MODULABS

- Led a beginner Python peer-learning team through weekly programming assignments; the team was selected as Outstanding Mission Team in Weeks 2, 4, 5, and 6.

Certifications

Nov. 2025	AI Utilization Analysis, Level 2	Korea Human Resource Development Institute (KRDI)
2025	Adobe Certified Professional: Photoshop, Premiere Pro, Illustrator, InDesign	Adobe

Skills

Programming & Data

- Python, JavaScript, OpenCV, Firebase, data preprocessing

AI & Interactive Systems

- OpenAI API, LLM-based interaction logic, text-to-speech (TTS) output integration

Hardware & Sensing

- Raspberry Pi, GPIO, camera-based image capture, Hall sensor arrays, NFC-triggered interaction, 3D-printed prototypes

Visualization & Design

- TouchDesigner, SketchUp, Adobe Creative Suite

Language

TEPS 366, 2026

TOEIC 895, 2026